

MTH 9821 — Variance reduction in a rough volatility model

Team 9: Jaskaran Kalra, William McDonnell

Equation numbers (1)–(21) refer to the assignment. Code: `src/`; every table and figure is produced by `python run_all.py` (see README).

Problem 1. Build and validate the simulator

(a)

Moments of Y_t . The integrand $s \mapsto \sqrt{2H}(t-s)^{H-1/2}$ is deterministic and square integrable on $[0, t]$ because $2H-1 > -1$. Hence Y_t is a Wiener integral: Gaussian with $\mathbb{E}[Y_t] = 0$, and by the Itô isometry

$$\text{Var}(Y_t) = 2H \int_0^t (t-s)^{2H-1} ds = 2H \cdot \frac{t^{2H}}{2H} = t^{2H}.$$

For $Z \sim N(0, v)$, $\mathbb{E}[e^{\eta Z}] = e^{\eta^2 v/2}$, so $\mathbb{E}[V_t] = \xi e^{-\eta^2 t^{2H}/2} \mathbb{E}[e^{\eta Y_t}] = \xi$. This is (3).

$a_H \in (0, 1)$. $a_H = \sqrt{2H}/(H + \frac{1}{2}) > 0$ for $H > 0$, and $a_H < 1 \iff 2H < (H + \frac{1}{2})^2 \iff 0 < (H - \frac{1}{2})^2$, which holds for $H \neq \frac{1}{2}$, in particular on $0 < H < \frac{1}{2}$. So $\sqrt{1 - a_H^2}$ is real.

Covariance with ΔW_j . With $\Delta W_j = \sqrt{\Delta} G_j$ and U_j independent of G_j ,

$$\text{Cov}\left(\Delta^H (a_H G_j + \sqrt{1 - a_H^2} U_j), \Delta W_j\right) = \Delta^H a_H \sqrt{\Delta} = \frac{\sqrt{2H}}{H + \frac{1}{2}} \Delta^{H+1/2}.$$

The exact most-recent-interval integral is $I_j = \sqrt{2H} \int_{t_{j-1}}^{t_j} (t_j - s)^{H-1/2} dW_s$. By the Itô isometry for the covariance of two Wiener integrals ($\Delta W_j = \int_{t_{j-1}}^{t_j} 1 dW_s$),

$$\text{Cov}(I_j, \Delta W_j) = \sqrt{2H} \int_{t_{j-1}}^{t_j} (t_j - s)^{H-1/2} ds = \frac{\sqrt{2H}}{H + \frac{1}{2}} \Delta^{H+1/2},$$

which matches. (Also $\text{Var}(I_j) = \Delta^{2H}$, equal to the variance $\Delta^{2H}(a_H^2 + 1 - a_H^2)$ of the first term of (9); both pairs are jointly Gaussian, so their joint laws agree.)

(b)

Implementation (`src/model.py`). w_k and q_j are precomputed. The sum in (9) is $\sum_{i=1}^{j-1} w_{j-i+1} \Delta W_i$, computed for all j at once as a product of the $(\text{paths} \times n)$ matrix of ΔW with a fixed upper-triangular $n \times n$ matrix $M_{ij} = w_{j-i+1} \mathbf{1}\{i < j\}$ (checked against a direct double loop to 10^{-15}). Then (11) gives $\hat{V}_j$, and (12) uses $\hat{V}_j$, $j = 0, \dots, n-1$, with $\Delta W_{j+1}, \Delta B_{j+1}$.

Derivation of (10). $\hat{Y}_j$ is a linear combination of the independent normals $G_1, \dots, G_j, U_j$, so it is Gaussian with mean zero. The first term involves only G_j, U_j ; the k -th term of the sum ($k \geq 2$) involves only G_{j-k+1} with $j-k+1 \leq j-1$. All terms are therefore independent, and

$$\text{Var}(\hat{Y}_j) = \Delta^{2H}(a_H^2 + 1 - a_H^2) + \sum_{k=2}^j w_k^2 \text{Var}(\Delta W_{j-k+1}) = \Delta^{2H} + \Delta \sum_{k=2}^j w_k^2 = q_j, \quad q_0 = 0.$$

Mean of $\widehat{V}_j$. Since $\widehat{Y}_j \sim N(0, q_j)$, $\mathbb{E}[\widehat{V}_j] = \xi e^{-\eta^2 q_j/2} \mathbb{E}[e^{\eta \widehat{Y}_j}] = \xi e^{-\eta^2 q_j/2} e^{\eta^2 q_j/2} = \xi$.

Replacing q_j by t_j^{2H} gives $\mathbb{E}[\widehat{V}_j] = \xi \exp(\frac{\eta^2}{2}(q_j - t_j^{2H}))$, which equals ξ only if $q_j = t_j^{2H}$. In general $q_j \neq t_j^{2H}$ because the kernel on earlier intervals is replaced by its average: by Cauchy–Schwarz, $\Delta w_k^2 = \frac{1}{\Delta} (\int g)^2 \leq \int g^2$ over each interval, with strict inequality since the kernel g is not constant there. Hence $q_j < t_j^{2H}$ for $j \geq 2$ (equality only at $j = 1$), and the discrete mean would fall below ξ . At $n = 128$, $q_n = 0.87010$ vs $T^{2H} = 0.87055$, a factor 0.9995 on $\mathbb{E}[\widehat{V}_n]$.

(c)

$n = 128$, baseline (5), 100,000 independent paths. Table 1.

j	t_j	$\overline{Y}_j$ (SE)	$s^2(\widehat{Y}_j)$	q_j	s^2/q_j	$\overline{V}_j$ (SE)
32	0.1250	+0.0016 (0.0026)	0.6574	0.6593	0.9971	0.04012 (0.00023)
64	0.2500	+0.0050 (0.0028)	0.7631	0.7574	1.0075	0.04054 (0.00027)
128	0.5000	-0.0012 (0.0030)	0.8772	0.8701	1.0082	0.04048 (0.00033)

Table 1: Sample moments of $\widehat{Y}_j$ and $\widehat{V}_j$; SE = $s/\sqrt{100,000}$. Targets: $\mathbb{E}\widehat{Y}_j = 0$, $\text{Var } \widehat{Y}_j = q_j$, $\mathbb{E}\widehat{V}_j = \xi = 0.04$.

All sample means of $\widehat{Y}_j$ are within 1.8 SE of zero; $\overline{V}_j - \xi$ is 0.5, 2.0 and 1.5 SE at $j = 32, 64, 128$ (the three j use the same paths, so these deviations are positively correlated). The variance ratios differ from one by at most 0.8%, i.e. at most 1.8 times the Gaussian relative standard error $\sqrt{2/N} = 0.45\%$ of a sample variance.

Figure 1 plots three variance paths generated from the same (G, U) inputs at $H = 0.1$ and $H = 0.4$. The $H = 0.1$ paths are much rougher, with sharp short-lived spikes; at $H = 0.4$ the same inputs give smoother paths with smaller excursions.

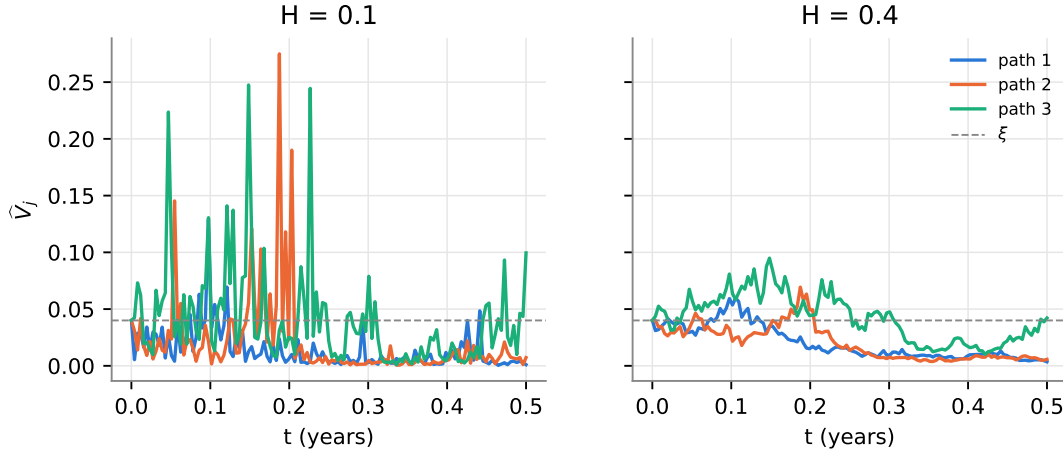


Figure 1: $\widehat{V}_j$, $n = 128$, $\xi = 0.04$, $\eta = 1.5$; same Gaussian inputs in both panels.

(d)

With $\eta = 0$, (11) gives $\widehat{V}_j = \xi$ for every j , whatever $\widehat{Y}_j$ and q_j are. Summing (12),

$$\log \widehat{S}_n = \log S_0 - \frac{1}{2}\xi T + \sqrt{\xi} \sum_{j=1}^n (\rho \Delta W_j + \sqrt{1 - \rho^2} \Delta B_j).$$

The sum is Gaussian with mean 0 and variance $n\Delta(\rho^2 + 1 - \rho^2) = T$, so $\log \widehat{S}_n \sim N(\log S_0 - \frac{1}{2}\xi T, \xi T)$ exactly, for every n : this is the Black–Scholes terminal law with zero rates and volatility $\sqrt{\xi}$. Hence $\mathbb{E}[(K - \widehat{S}_n)^+] =$

$\text{BS}_{\text{put}}(S_0, K, \xi T)$. Table 2: 100,000 paths per n , independent streams for each n ; all $|z| \leq 2.02$.

n	K	MC price	SE	BS (6)	z
1	90	1.7776	0.0131	1.7725	+0.39
1	100	5.6379	0.0241	5.6372	+0.03
1	110	12.2070	0.0339	12.2112	-0.12
8	90	1.7869	0.0131	1.7725	+1.10
8	100	5.6620	0.0241	5.6372	+1.03
8	110	12.2381	0.0340	12.2112	+0.79
128	90	1.7991	0.0132	1.7725	+2.02
128	100	5.6787	0.0242	5.6372	+1.71
128	110	12.2482	0.0341	12.2112	+1.09

Table 2: $\eta = 0$: simulated put prices vs (6) with $a = \xi T$; $z = (\text{MC} - \text{BS})/\text{SE}$. The three strikes at a given n share paths.

A bug these checks detect: driving ΔB with the same normals as ΔW (a dependence bug). The return variance becomes $(\rho + \sqrt{1 - \rho^2})^2 T \neq T$ and the prices move far from (6). Wrong increment scaling or a missing $-\frac{1}{2}\hat{V}_j\Delta$ term is caught the same way.

A bug they do not detect: anything in the variance driver, e.g. an off-by-one in the convolution index (ΔW_{j-k+2} instead of ΔW_{j-k+1}) or look-ahead (using $\hat{V}_{j+1}$ in (12)). At $\eta = 0$ the variance path is constant, so these errors have no effect.

Problem 2. A Black–Scholes control variate

(a)

By the argument of 1(d), $\log \tilde{S}_T \sim N(\log S_0 - \frac{1}{2}\xi T, \xi T)$, so

$$c_K = \mathbb{E}[C_K] = \text{BS}_{\text{put}}(S_0, K, \xi T).$$

For fixed β : $\mathbb{E}[R] = \mathbb{E}[X] - \beta(\mathbb{E}[C] - c_K) = p_{K,n}$. Its variance is $\text{Var}(R) = \text{Var} X - 2\beta \text{Cov}(X, C) + \beta^2 \text{Var} C$, a quadratic in β minimised at

$$\beta^* = \frac{\text{Cov}(X, C)}{\text{Var} C}, \quad \text{Var}(R^*) = \text{Var} X (1 - \text{Corr}(X, C)^2).$$

If $\tilde{S}_T$ were simulated independently, $\text{Cov}(X, C) = 0$, so $\beta^* = 0$ and the control gives no reduction; any $\beta \neq 0$ adds $\beta^2 \text{Var} C$. The control only helps through its correlation with X , which comes from sharing the increments $\Delta W_j, \Delta B_j$.

(b)

$\hat{\beta}_K = \widehat{\text{Cov}}(X, C)/\widehat{\text{Var}}(C)$ is computed from an independent pilot and frozen. Conditional on the pilot, $\hat{\beta}_K$ is a constant and the production R_i are i.i.d. with $\mathbb{E}[R_i \mid \text{pilot}] = p_{K,n} - \hat{\beta}_K(\mathbb{E}[C] - c_K) = p_{K,n}$. So the estimator is unbiased conditional on the pilot, and (18) is the correct SE, because the R_i are i.i.d. given the pilot.

If $\hat{\beta}$ is fitted on the production sample, it depends on the same C_i , and $\mathbb{E}[\hat{\beta}(\bar{C} - c_K)] \neq 0$ in general: the estimator has an $O(1/m)$ bias. The R_i are also no longer independent (all share $\hat{\beta}$), so (18) is justified only asymptotically.

Zero pilot control variance. If the pilot sample variance of C is zero (e.g. all pilot controls equal zero), the code sets $\hat{\beta} = 0$, so the estimator reduces to plain Monte Carlo (`_coef` in `src/estimators.py`). This did not occur in any run.

(c)

Implemented in `control.variate()`: pilot of 10,000 paths, then production $R_i = X_i - \hat{\beta}(C_i - c_K)$ with SE and the 95% interval from (18), using production observations only.

Pathwise identity at $\eta = 0$. Then $\hat{V}_j \equiv \xi$, and (12) and (14) give the same $\log \hat{S}_n = \log \tilde{S}_T$, so $X = C$ on every path and, with $\beta = 1$, $R = c_K$. Table 3 (10,000 paths): $\max_i |R_i - c_K| \leq 10^{-13}$ (roundoff; the two logs are accumulated in different orders, $\max |\hat{S}_n / \tilde{S}_T - 1| = 1.1 \times 10^{-15}$).

K	c_K	$\max_i R_i - c_K $	$s(R)$
90	1.772451	8.5e-14	1.6e-14
100	5.637198	9.9e-14	2.4e-14
110	12.211246	9.9e-14	3.0e-14

Table 3: $\eta = 0$, $\beta = 1$: adjusted payoffs are constant up to roundoff.

Why an estimated coefficient need not reduce variance. From (a), $\text{Var}(R(\beta)) - \text{Var}(X) = \text{Var}(C)[(\beta - \beta^*)^2 - \beta^{*2}]$, so a fixed β reduces variance only if $|\beta - \beta^*| < |\beta^*|$. The pilot $\hat{\beta}$ is random and can fall outside this range, especially with a small pilot or weak correlation. Even when it is inside, the reduction is a statement about population variances: in a given finite production sample the sample variance of R can still exceed that of X by chance.

Problem 3. Antithetic sampling

(a)

$Z = (G, U, D)$ is a vector of i.i.d. $N(0, 1)$ entries, and the standard normal law is symmetric, so $-Z \stackrel{d}{=} Z$. $X^\pm$ are the same measurable function of $\pm Z$, so $X^- \stackrel{d}{=} X^+$, the correct law. $\hat{Y}_j$ is linear in (G, U) , so negation gives $-\hat{Y}_j$, and

$$\hat{V}_j^+ = \xi \exp(\eta \hat{Y}_j - \frac{\eta^2}{2} q_j), \quad \hat{V}_j^- = \xi \exp(-\eta \hat{Y}_j - \frac{\eta^2}{2} q_j).$$

Negating only the stock-return increments while keeping $\hat{V}^+$ drives the second stock with $-\Delta W$ but a variance path built from $+\Delta W$. The pair (stock driver, variance driver) then has correlation $-\rho$ instead of ρ , so the second path does not have the model's law (for $\rho \neq 0$). It is not the full reflection $Z \mapsto -Z$, under which the variance path changes too.

(b)

Let $\sigma^2 = \text{Var } X^+ = \text{Var } X^-$ and $r = \text{Corr}(X^+, X^-)$. For M i.i.d. pair averages,

$$\text{Var}(\bar{A}_M) = \frac{\text{Var}(A)}{M} = \frac{\sigma^2 + \sigma^2 + 2r\sigma^2}{4M} = \frac{\sigma^2(1+r)}{2M}, \quad \text{Var}(\bar{X}_{2M}) = \frac{\sigma^2}{2M},$$

$$\frac{\text{Var}(\bar{A}_M)}{\text{Var}(\bar{X}_{2M})} = 1 + \text{Corr}(X^+, X^-).$$

Antithetic sampling reduces variance iff $\text{Corr}(X^+, X^-) < 0$. This is *not* guaranteed here. The guarantee holds when the payoff is a monotone function of each input, and $Z \mapsto X$ is not. Raising G_j moves the stock through the $\rho \Delta W$ term (down, since $\rho < 0$) and also raises later variances through $\hat{Y}$ (exponentially), which changes both the drift $-\frac{1}{2}\hat{V}\Delta$ and the size of subsequent returns. The put payoff also has a kink. With the even (variance-driven) part of X dominant, e.g. large η and a far out-of-the-money strike, r can be close to zero or positive.

(c)

Implemented in `antithetic()`: each batch draws one (G, U, D) per pair and simulates Z and $-Z$. The statistic is the pair average $A_i, i = 1, \dots, M$; $\widehat{SE} = \sqrt{s_A^2/M}$ and the 95% interval follow (18) with $m = M$.

Treating the $2M$ payoffs as independent uses $s_X^2/(2M)$, which estimates $\sigma^2/(2M)$ and omits the within-pair covariance term $2r\sigma^2/(4M)$. The true variance is $\sigma^2(1+r)/(2M)$. The naive SE is therefore off by the factor $\sqrt{1+r}$: too large when $r < 0$, too small when $r > 0$. Only the pairs are independent sampling units.

Problem 4. Combining the two methods

(a)

$\text{Var}(R_{\text{pair}}) = \text{Var } A - 2\gamma \text{Cov}(A, C_A) + \gamma^2 \text{Var } C_A$, minimised at

$$\gamma^* = \frac{\text{Cov}(A, C_A)}{\text{Var}(C_A)}, \quad \text{Var}(R_{\text{pair}}^*) = \text{Var } A (1 - \text{Corr}(A, C_A)^2).$$

Write each payoff as even plus odd parts in Z : $X = X_e + X_o$ with $X_e(Z) = \frac{1}{2}(X(Z) + X(-Z))$, $X_o(Z) = \frac{1}{2}(X(Z) - X(-Z))$, and likewise C . Since $Z \stackrel{d}{=} -Z$, even and odd functions are uncorrelated, so

$$\beta^* = \frac{\text{Cov}(X_e, C_e) + \text{Cov}(X_o, C_o)}{\text{Var } C_e + \text{Var } C_o}, \quad \gamma^* = \frac{\text{Cov}(X_e, C_e)}{\text{Var } C_e} \quad (A = X_e, C_A = C_e).$$

β^* is a variance-weighted mix of the even and odd regression slopes. γ^* is the even slope alone, because pairing has already removed the odd parts. They coincide only if the two slopes are equal.

(b)

Implemented in `combined()`: pilot of 5,000 independent pairs gives $\hat{\gamma}$, frozen for $M = 50,000$ production pairs; $\widehat{SE} = \sqrt{s_{R_{\text{pair}}}^2/M}$, where $s_{R_{\text{pair}}}^2$ is the sample variance of the M production values $R_{\text{pair},i}$.

Unbiasedness given the pilot. Each $C(\pm Z)$ has the law of C , so $\mathbb{E}[C_A] = c_K$. Given the pilot, $\hat{\gamma}$ is constant and $\mathbb{E}[R_{\text{pair}} \mid \text{pilot}] = \mathbb{E}[A] - \hat{\gamma}(\mathbb{E}[C_A] - c_K) = p_{K,n}$. Numerically, the production mean of C_A differs from c_K by 1.11, 0.99, 1.16 SE at $K = 90, 100, 110$, consistent with $\mathbb{E}[C_A] = c_K$. Every combined estimate lies inside the plain-MC 95% interval (Table 4).

Problem 5. Numerical investigation

(a) **Baseline comparison**

Parameters (5), $n = 128$; 100,000 production path evaluations per method (paired methods: 50,000 pairs); pilots of 10,000 paths (CV) and 5,000 pairs (AV+CV). Each method uses its own independent streams, and all three strikes are evaluated on the same paths within a method.

K	Method	Price	SE	95% CI	m
90	Plain	1.9849	0.0184	[1.9489, 2.0209]	100,000
90	CV	2.0221	0.0114	[1.9997, 2.0445]	100,000
90	AV	2.0212	0.0175	[1.9869, 2.0554]	50,000
90	AV+CV	2.0142	0.0115	[1.9917, 2.0367]	50,000
100	Plain	5.0614	0.0280	[5.0065, 5.1163]	100,000
100	CV	5.1082	0.0157	[5.0774, 5.1391]	100,000
100	AV	5.0951	0.0239	[5.0482, 5.1419]	50,000
100	AV+CV	5.1060	0.0162	[5.0743, 5.1378]	50,000
110	Plain	11.2223	0.0368	[11.1503, 11.2944]	100,000
110	CV	11.2684	0.0194	[11.2304, 11.3064]	100,000
110	AV	11.2406	0.0270	[11.1878, 11.2934]	50,000
110	AV+CV	11.2449	0.0222	[11.2014, 11.2884]	50,000

Table 4: Estimates, SE and 95% intervals from (18); m = production observations (paths or pairs).

Method	pilot evals	prod. evals	pilot (s)	prod. (s)	total (s)	total range (s)
Plain	0	100,000	0.000	1.226	1.226	1.176–1.318
CV	10,000	100,000	0.153	1.523	1.676	1.565–1.882
AV	0	100,000	0.000	0.906	0.906	0.883–0.930
AV+CV	10,000	100,000	0.116	1.187	1.302	1.235–1.338

Table 5: Computation time per method, shared by all three strikes (counted once). Pilot and production times are medians of 5 repeated runs with identical seeds (identical estimates); the last column is the range of the 5 totals. Timing follows the protocol: it includes RNG, simulation, control evaluation, coefficient estimation and production statistics, and excludes plotting, file output and a one-time warm-up.

K	Method	coef.	$\hat{\rho}_{X,C}$	$\hat{\rho}_{X^+,X^-}$	$\hat{\rho}_{A,C_A}$	VRF	VRF _{pred}	Gain
90	Plain	–	–	–	–	1.00	1.00	1.00
90	CV	1.1420	0.7900	–	–	2.59	2.66	1.90
90	AV	–	–	-0.1164	–	1.10	1.13	1.49
90	AV+CV	1.1575	–	-0.1146	0.7599	2.56	2.67	2.41
100	Plain	–	–	–	–	1.00	1.00	1.00
100	CV	0.9842	0.8302	–	–	3.16	3.22	2.32
100	AV	–	–	-0.2838	–	1.37	1.40	1.86
100	AV+CV	1.1149	–	-0.2836	0.7387	2.99	3.07	2.82
110	Plain	–	–	–	–	1.00	1.00	1.00
110	CV	0.9333	0.8515	–	–	3.59	3.64	2.63
110	AV	–	–	-0.4699	–	1.86	1.89	2.52
110	AV+CV	1.4790	–	-0.4671	0.5759	2.75	2.81	2.59

Table 6: Fitted coefficients ($\hat{\beta}$ for CV, $\hat{\gamma}$ for AV+CV, from the pilot), production-sample correlations, VRF and Gain from (19) using the total times of Table 5. VRF_{pred} is the large-sample value implied by the correlations: $1/(1 - \rho_{X,C}^2)$, $1/(1 + \rho_{X^+,X^-})$, $1/[(1 + \rho_{X^+,X^-})(1 - \rho_{A,C_A}^2)]$.

(b) Explanation

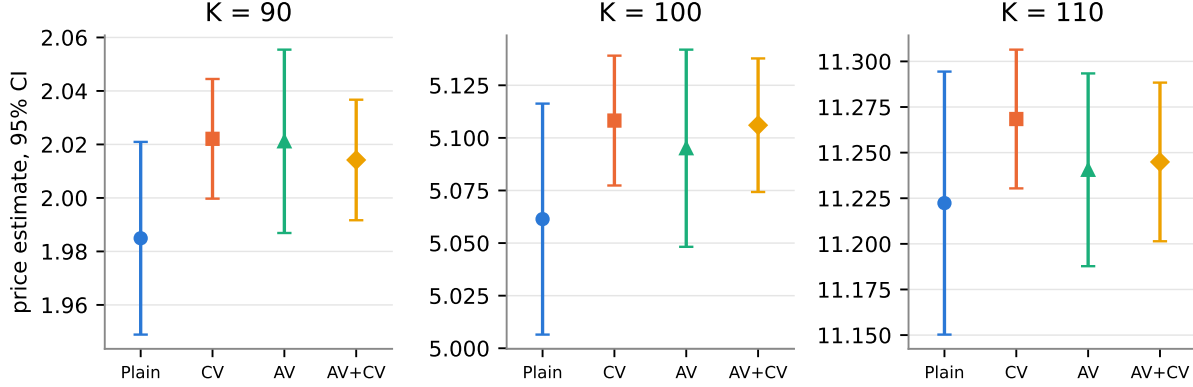


Figure 2: 95% confidence intervals of the four estimators, baseline parameters (5), $n = 128$.

All intervals overlap at every strike (Figure 2). Plain MC sits below the others at all three strikes; its three estimates share paths, so their errors are positively correlated.

Control variate. $\rho_{X,C} = 0.79, 0.83, 0.85$ at $K = 90, 100, 110$, giving observed VRF 2.59, 3.16, 3.59, close to $1/(1 - \rho^2)$. The control uses the same Brownian increments with variance frozen at ξ , so it tracks the stock-noise part of X . The correlation rises with K : the in-the-money put is closer to linear in S_T , while the out-of-the-money put depends more on the tails, where the rough variance path matters most.

Antithetic. $\rho_{X^+, X^-} = -0.12, -0.28, -0.47$, giving VRF 1.10, 1.37, 1.86 $\approx 1/(1 + \rho)$. Reflection cancels the odd part of X in Z ; the payoff is closer to odd (linear) in Z in the money, so the gain grows with K .

Combination. VRF 2.56, 2.99, 2.75: better than AV alone, but slightly *below* CV alone at every strike. Pairing removes the odd part of X , and that is mostly what the control also explains (the control is driven by the same increments). The control is left to explain the even part, which it does worse: $\rho_{A, C_A} = 0.76, 0.74, 0.58 < \rho_{X, C}$. Hence $(1 + \rho_{X^+, X^-})(1 - \rho_{A, C_A}^2)$ is not smaller than $1 - \rho_{X, C}^2$; the two reductions overlap rather than multiply. The fitted $\hat{\gamma} \neq \hat{\beta}$ (e.g. 1.48 vs 0.93 at $K = 110$), as 4(a) predicts.

Efficiency and ranking. By VRF: $CV > AV+CV > AV$ at every strike. By Gain the order changes, because cost per path evaluation differs. AV draws one set of normals for two paths and needs no control, so it is cheapest (0.91s vs 1.23s for plain). CV evaluates the control and runs a pilot (1.68s). AV+CV saves on random numbers and pays for the control (1.30s). Hence Gain is $\{CV\ 1.90, AV\ 1.49, AV+CV\ 2.41\}$ at $K = 90$, $\{2.32, 1.86, 2.82\}$ at $K = 100$, and $\{2.63, 2.52, 2.59\}$ at $K = 110$. **The most efficient method by (19) is AV+CV at $K = 90$ and $K = 100$.** At $K = 110$ the three Gains differ by at most 4%, comparable to the run-to-run timing variability (Table 5), so no method is clearly most efficient there. The disagreement between VRF and Gain rankings comes entirely from random-number generation cost: in this NumPy implementation it is a large share of the work, and antithetic sampling halves it. A different implementation or hardware would change the time ratios but not the VRFs.

(c) Parameter sensitivity ($K = 100$, $n = 128$)

100,000 production path evaluations per method, pilots as in (a), one parameter changed at a time.

Case	Method	Price	SE	coef.	$\hat{\rho}_{X,C}$	$\hat{\rho}_{X^+,X^-}$	$\hat{\rho}_{A,C_A}$	VRF	Gain
$\eta = 0.5$	Plain	5.5478	0.0259	–	–	–	–	1.00	1.00
$\eta = 0.5$	CV	5.5536	0.0057	1.0434	0.9749	–	–	20.37	14.58
$\eta = 0.5$	AV	5.5573	0.0189	–	–	-0.4609	–	1.87	2.66
$\eta = 0.5$	AV+CV	5.5536	0.0056	1.1231	–	-0.4577	0.9550	21.05	19.03
$H = 0.4$	Plain	5.1779	0.0283	–	–	–	–	1.00	1.00
$H = 0.4$	CV	5.1434	0.0106	1.0834	0.9252	–	–	7.09	5.38
$H = 0.4$	AV	5.1641	0.0231	–	–	-0.3322	–	1.51	2.28
$H = 0.4$	AV+CV	5.1470	0.0096	1.2782	–	-0.3358	0.9061	8.62	8.99

Table 7: $K = 100$; coef. is $\hat{\beta}$ (CV) or $\hat{\gamma}$ (AV+CV). Baseline values at $K = 100$: VRF CV 3.16, AV 1.37, AV+CV 2.99; Gain 2.32, 1.86, 2.82.

$\eta = 0.5$. Weaker variance fluctuations make $\hat{S}_n$ close to the Black–Scholes control: $\rho_{X,C}$ rises from 0.83 to 0.975 and the CV VRF from 3.16 to 20.4. The pair correlation also becomes more negative ($-0.28 \rightarrow -0.46$), since the payoff is closer to odd in Z . After pairing, the control still explains the even part well ($\rho_{A,C_A} = 0.955$), so the combination now beats each method separately (VRF 21.1, Gain 19.0, the most efficient).

$H = 0.4$. Smoother variance paths give a milder improvement: $\rho_{X,C} = 0.925$ (CV VRF 7.1), $\rho_{X^+,X^-} = -0.33$ (AV VRF 1.5) and $\rho_{A,C_A} = 0.906$. The combination again improves on both (VRF 8.6, Gain 9.0, the most efficient). In both cases the relative timing of methods is unchanged from the baseline (Gain $\approx$ VRF $\times$ cost ratio), so the efficiency changes come from the correlations.

(d) Reproducibility and implementation

- **Software/hardware:** Python 3.11.15, NumPy 2.4.4, SciPy 1.17.1 (Matplotlib for figures); Linux x86_64, Intel Xeon @ 2.80 GHz, 2 logical CPUs; float64 throughout (`results/environment.json`).
- **Seeds:** every stream is `PCG64(SeedSequence(9821, spawn_key=key))`. Keys: 1(c) moments (1, 3), 1(c) plot (1, 4), 1(d) (1, 5, i) for the i -th n , 2(c) (1, 6); Problem 5(a) (5, method, stage) with method $\in \{0, 1, 2, 3\}$ = Plain, CV, AV, AV+CV and stage 0 = pilot, 1 = production; 5(c) (60, $\cdot$, $\cdot$) for $\eta = 0.5$ and (61, $\cdot$, $\cdot$) for $H = 0.4$; Bonus (7, 0, 1). Within a batch the normals are drawn as G , then U , then D , each of shape (batch, n).
- **Batching:** 10,000 path evaluations per batch (5,000 pairs for paired methods). Means and co-moment matrices are combined across batches with the pairwise (Chan et al.) update; no path arrays are stored.
- **Pilot vs production:** each method has distinct spawn keys for pilot and production, so the two streams are independent, and the coefficient is computed and frozen before any production draw.
- **Pairs as units:** one row of (G, U, D) is drawn per pair; the two members are Z and $-Z$ of that row. Statistics are accumulated on the pair average (or adjusted pair average), so each pair contributes one observation, and $m = M$ in (18).

Bonus. Conditional Monte Carlo

(a)

Let $\mathcal{G} = \sigma(G_1, \dots, G_n, U_1, \dots, U_n)$. Then $\hat{V}_0, \dots, \hat{V}_{n-1}$, I_n and J_n are $\mathcal{G}$ -measurable, and summing (12) gives

$$\log \hat{S}_n = \log S_0 - \frac{1}{2} I_n + \rho J_n + \sqrt{1 - \rho^2} \sum_{j=0}^{n-1} \sqrt{\hat{V}_j} \Delta B_{j+1}.$$

The ΔB are independent of $\mathcal{G}$, so given $\mathcal{G}$ the last sum is $N(0, I_n)$, and

$$\log \hat{S}_n \mid \mathcal{G} \sim N\left(\log S_0 + \rho J_n - \frac{1}{2} I_n, (1 - \rho^2) I_n\right).$$

With $s^* = S_0 \exp(\rho J_n - \frac{\rho^2}{2} I_n)$ and $a = (1 - \rho^2) I_n$, the mean is $\log s^* - \frac{1}{2}a$. This is the Black–Scholes law with spot s^* and total variance a , so $Q_K = \mathbb{E}[(K - \hat{S}_n)^+ | \mathcal{G}] = \text{BS}_{\text{put}}(s^*, K, a)$, which is (21). Using S_0 as the spot ignores the $\mathcal{G}$ -measurable shift $\rho J_n - \frac{\rho^2}{2} I_n$, i.e. the part of the return driven by the same W as the variance. It is correct only if $\rho = 0$.

(b)

By the law of total variance, $\text{Var } X_K^{(n)} = \mathbb{E}[\text{Var}(X_K^{(n)} | \mathcal{G})] + \text{Var}(Q_K) \geq \text{Var}(Q_K)$, with the gap equal to the average conditional variance from the D noise. Reflecting only D (keeping G, U) gives the pair average $\frac{1}{2}(X(G, U, D) + X(G, U, -D))$. Its conditional mean is also Q_K , and it removes the part of the D -noise that is odd in D . Exact conditional averaging integrates over D completely and removes all of it; D -reflection is a two-point approximation of that integral.

(c)

100,000 independent (G, U) samples, baseline (5), $K = 100$, $n = 128$; time is the median of 5 runs.

Method	Price	SE	time (s)	VRF	Gain
Plain	5.0614	0.0280	1.226	1.00	1.00
CV	5.1082	0.0157	1.676	3.16	2.32
AV	5.0951	0.0239	0.906	1.37	1.86
AV+CV	5.1060	0.0162	1.302	2.99	2.82
Conditional	5.1362	0.0211	0.799	1.76	2.70

Table 8: $K = 100$: conditional estimator vs the four methods of 5(a); VRF and Gain relative to plain MC.

The conditional estimator has VRF 1.76 and Gain 2.70, against Gains 2.32 (CV), 1.86 (AV) and 2.82 (AV+CV): better than AV, below AV+CV. Its VRF is limited because it removes only the independent stock noise (weight $1 - \rho^2 = 0.51$); the variance-path and ρJ_n noise remain.

Cost per sample. One sample needs no D draws (two thirds of the normals) and no stock recursion (12). It needs the variance path, two sums I_n, J_n , and one Black–Scholes formula per strike, so it is cheaper than a full stock path (0.80 s vs 1.23 s per 100,000).